

e# Sample spim program

#

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```
.data
prompt: .asciiz "Enter in an integer: "
str1:   .asciiz "the answer is: "
newline: .asciiz "\n"
bye:    .asciiz "Goodbye!\n"
##      .globl  main
```

```
.text
main:
```

```
# initialize
li      $s0, 10

# prompt for input
li      $v0, 4
la      $a0, prompt
syscall

# read in the value
li      $v0, 5
syscall
move     $s0, $v0
```

loop:

```
# print str1
li      $v0, 4
la      $a0, str1
syscall
```

```
# print loop value
li      $v0, 1
move     $a0, $s0
syscall
```

```
# print newline
li      $v0, 4
la      $a0, newline
syscall
```

```
# decrement loop value and branch if not negative
sub      $s0, $s0, 1
bgez     $s0, loop
```

```
# print goodbye message
li      $v0, 4
la      $a0, bye
```

```
syscall
jr $ra
```